

# A Theory of Bounded Observation

New theories, not just new theorems. The record-thermodynamic formalism: state, potentials, laws, phases, dynamics, and a measurement protocol for the bounded observer — held to its three first experimental confrontations. The record-phase p-scan localizes the first boundary at 0.233810 exactly; the rebuilt three-replica chain walks the second boundary to 0.305; the E. coli cycle scan, resolved by layer, separates the epsilon-laws from the k-laws and certifies 0.361 per cycle as the cascade's slowest case; and the optic-Nehari lemma is refuted exactly, repaired stably, and **Aring Abae** the small-gain law that the whole programme converges on.

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# 1 · The Mandate: Theories, Not Just Theorems

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The instruction that governs this volume is an upgrade of standard, and it should be quoted before it is interpreted: we want new theories, not just new theorems — new physics, new laws, enhanced understanding, substantial and conceptually novel advances in understanding or in technological capability. Volume III answered the previous form of the mandate with laws and theorems, each with its proved half separated from its open core. A pile of true statements, however well certified, is not yet a theory, and the difference is structural rather than rhetorical. A theorem closes a question; a theory opens a family of them. A theory fixes state variables and control variables, writes equations that relate them, and then returns predictions for experiments nobody has yet run — which is exactly what a pile of theorems cannot do, because theorems do not interpolate.

Thermodynamics is the controlling precedent, and it was already invoked in Volume III at the moment of legislation. Before Carnot and Clausius, the science of engines was exactly a pile: no engine exceeds Carnot efficiency, no process un-mixes heat, no perpetual motion machine works — each locally true, none deducible from the others. The four laws legislated the pile, and the legislation had a specific grammatical effect: the facts became *equations of state*, the engines became *systems* with state variables, and the design of new engines became *computation* on the formalism rather than trial against the pile. That is the transformation this volume owes the corpus: the bounded observer must stop being a collection of no-go theorems and become a system — with a state, controls, potentials, a phase diagram, a dynamics, and a measurement protocol.

The second thing the mandate demands, and the thing that separates this volume from a manifesto, is confrontation. A theory that has never met a number it did not already contain is a formalism, not a physics. The programme was left, at the end of Volume III, with a priority queue whose first three orders were cheap because the apparatus already existed: run the record-phase scan; run the E. coli cycle-size scan; state and attack optic-Nehari. All three orders have now been executed, and each returned numbers the corpus did not contain — a kink localized to five decimals, a second boundary approached by a rebuilt chain, a layer-resolved verdict that refutes one reading of a law while confirming another. The theory in this volume is built *with* those numbers, not beside them: its state variables are chosen so that the three confrontations become measurements of its sectors.

*A theorem closes a question. A theory opens a family of them, and then predicts the family's members before they are measured.*

## 2 · The Formalism: Record Thermodynamics of Bounded Observers

The theory — call it **record thermodynamics of bounded observers** — takes as its primitive the same primitive the whole programme settled on in Volume I: a behaviour factors through what its observer can distinguish. The **state** of a bounded observer is a pair  $(O, R)$ : the obstruction datum  $O$  — the spectra, widths, and intercept defects that encode what the observer cannot distinguish — and the record statistics  $R$  — the large-deviation family of what the observer actually registers. The **control variables** are the knobs the experimenter turns: the monitoring rate  $p$  in the quantum sector, the width and memory budget  $M$  in the compositional sector, and the loop size  $\epsilon$  and cycle index  $k$  in the cellular sector. Every object the corpus ever defined is either a state component, a control, a function of the two, or a theorem about how the functions respond to the knobs — and that is the claim of Table 1, checked line by line against the corpus's own inventory.

The **potentials** are the free-energy surfaces over the controls. The record scaled cumulant generating function  $\Lambda(\beta; p)$  is the primary one: its annealed limits are computable in exact arithmetic, its  $\beta = 1$  endpoint is the two-replica partition function — exactly the Ising model solved by Houtappel — and its  $\beta = 2$  endpoint is the three-replica partition function through the Clifford conjugation 3-design identity. Around it sit the derived thermodynamic quantities: the ESS budget functional, which plays the role of a dissipation inequality for sampling; the annealed-quenched gap, which is the interaction functional between the disorder and the record; and the replica interpolation identity, which monotonizes the gap exactly the way a thermodynamic inequality should. The theory's postulates are four, each with a status: P1, potential existence (proved for the circuit family); P2, replica closure (proved, the Z3-verifiable identities); P3, self-averaging (measured, the 0.105-per-site variance that is  $L$ -independent); P4, the boundary condition — phase boundaries occur at replica-eigenvalue crossings (proved in the two-level fragment, and now measured at two replica numbers).

The **phase structure** is the theory's newest organ and the one this session did the most to make physical. The annealed replica chain carries a ladder of non-analytic points — the record phases — at  $p_c^{(2)} = 0.233810$  and its successors 0.305(3), approximately 0.383, and 0.47 to 0.48, with the  $qu = n$  Potts classes predicted along the chain and the first-order character confirmed at  $n = 5$ . The **dynamics** is the cascade: composite updates with per-optic Lipschitz constants, geometric relaxation at rate minus the log of their product, and — new this session — a pricing rule for correlation defects, the in-span/transverse dichotomy of Theorem VIII. The **measurement protocol** is anchor-reproduction-first: no confrontational claim is admitted until the apparatus has reproduced the corpus's own certified numbers, after which the new numbers inherit the same error contracts. Table 1 fixes all of this in one place, and Figure 1 draws the stack.

| Sector             | Object                    | Content                                                                                                                                                          | Status                                                                                                            |
|--------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| State              | Obstruction datum O       | Hankel spectra, flat widths, intercept defects — the observer's conserved ignorance; transported and priced, never destroyed (Law 1).                            | PROVED-CORPUS (transport, intercept); the conservation reading is Theorem I                                       |
| State              | Record statistics R(beta) | the Born record's large-deviation family: SCGF, rate function, multifractal spectrum, self-averaging variance 0.105/site (L-independent).                        | PROVED-CORPUS (definitions, no-freeze); MEASURED (variance, annealed-quenched gap 0.0226 nats/site)               |
| Controls           | (p, M, budget)            | monitoring rate p (quantum sector); width and memory budget M (compositional sector); loop size epsilon and cycle index k (cellular sector).                     | DEFINITION; each substrate fixes its own control geometry                                                         |
| Potentials         | Lambda(beta; p)           | the record free energy: annealed replica chain, exact closure at beta = 1 (two-replica Ising) and beta = 2 (three-replica, Clifford 3-design).                   | PROVED-CORPUS (closure identities); the potential picture is this volume's framing                                |
| Potentials         | ESS budget                | $\exp[-2Lt(\text{Lambda}(2) - 2 \text{Lambda}(1))]$ : the dissipation inequality of sampling; the budget curve is exact at the production cells.                 | PROVED-CORPUS (identity); MEASURED (0.0340/0.0349/0.0353 at p = 0.16)                                             |
| Equations of state | Four laws + response laws | Laws 0-3 of Volume III, now read as equations of state; the response laws: inverse relative-gap concentration, Onsager log growth, the drift law.                | MIXED (see Table 4 of the ledger); response laws MEASURED-HERE at n = 2                                           |
| Phase structure    | p_c chain                 | 0.233810 (exact) < 0.305(3) < ~0.383 < 0.47-0.48: record phases at replica-eigenvalue crossings; annealed points recede from the quenched 0.1597(8).             | PROVED (n = 2 closed form); MEASURED-HERE (n = 2 kink; n = 3 ladder); PROVED-COMPUTATIONAL (n = 4, 5)             |
| Dynamics           | Cascade + defect pricing  | geometric relaxation at rate $-\ln(\text{Prod } L)$ per cycle; defects priced by the in-span/transverse dichotomy; accumulation obeys the small-gain inequality. | PROVED-CORPUS (Banach structure); PROVED-HERE (dichotomy, Theorem VIII); MEASURED-HERE (layers, cascade envelope) |
| Measurement        | Three confrontations      | the p-scan (n = 2), the Lambda(2) chain (n = 3), the E. coli layer-resolved cycle scan — each with an anchor-reproduction gate before any new claim.             | RUN-HERE; verdicts in Tables 2 and 3                                                                              |

Table 1 — The formalism at a glance: objects, status, and where each ingredient lives

## THE STATE

**O Obstruction datum**

what the observer cannot distinguish: spectra, widths, intercepts — the conserved charge

**R Record statistics**

what the observer does register: the Born record with SCGF  $\Lambda(\beta, p)$  as its potential

## THE POTENTIALS

 **$\Lambda$  Free-energy surfaces over the control variables (p, M, budget)**

replica chain  $\lambda_1(p) \rightarrow$  Houtappel free energy; ESS budget  $\exp[-2\text{Lt}(\Lambda(2)-2\Lambda(1))]$ ; annealed-quenched gap as the interaction functional

## THE LAWS (EQUATIONS OF STATE)

**0 · Equilibrium blindness**

at equilibrium the obstruction is inert; no inside arrow of time

**1 · Conservation**

ignorance transports, relocates, multiplies — never destroyed

**2 · Distortion production**

nonzero obstruction produces distortion; Cesaro floor = AAK tail

**3 · Unattainability**

zero obstruction at finite price unreachable; one wall

## THE PHASE STRUCTURE (THIS SESSION: THE FIRST TWO BOUNDARIES MEASURED)

**0.233810** $p_c^{(2)} \cdot n=2$  exact**0.305(3)** $p_c^{(3)} \cdot \Lambda(2)$  kink **$\approx 0.383$**  $p_c^{(4)}$ **0.47–0.48** $p_c^{(5)}$  · first order

record phases of the monitored circuit; annealed points recede from the quenched 0.1597(8)

## THE DYNAMICS

 **$\Pi L$  Cascade contraction**

geometric relaxation, rate  $-\ln(\Pi L)$ ; E. coli certified slowest case 0.361/cycle

 **$\Delta$  Defect pricing**

in-span defects price the factorization quadratically; transverse defects price the value linearly

## THE MEASUREMENTS — THREE CONFRONTATIONS, THIS SESSION

**I · The p-scan (n = 2)**

kink localized at 0.23381 exactly; Onsager law  $R^2 = 0.99$ ; gap closes 0.942/L

**CONFIRMED · every anchor reproduced**

**II · The  $\Lambda(2)$  chain (n = 3)**

Weingarten channel rebuilt; ladder 0.271  $\rightarrow$  0.297  $\rightarrow$  0.305(3); anchors exact to 6 decimals

**CONFIRMED · first ladder to the second boundary**

**III · E. coli, layer-resolved**

genotype layer: one-pass saturation; parameter layer 0.249, drift law 1.001; cascade 0.361 slowest case

**LAYER-RESOLVED · the k-laws live above the LP layer**

Figure 1 — The theory stack: state, potentials, laws, phase structure, dynamics, and the measurement layer. The first two phase boundaries and the three confrontations are this session's entries; every number in the measurement band is either exact or carries its fit quality.

## 3 · Confrontation I: The Record-Phase p-Scan at $n = 2$

The prediction being tested is Volume III's headline physics: the Born records of monitored quantum circuits carry a phase structure, and the first boundary of that structure is the certified point 0.233810. The apparatus is the manuscript's own exact solution — the bond channel identified with the anisotropic triangular-lattice Ising model, the compressed operator similar to a positive semidefinite matrix, the complete spectrum in Kaufman's free-fermion form, and Houtappel's closed-form free energy. The implementation reproduced every published anchor before any new number was taken: the finite-size crossing table (0.23319, 0.23348, 0.23361, 0.23368) to six decimals; the growth rate  $\lambda_1(28)^{1/28} = 0.683844946$  at  $p = 0.40$  to nine digits; the amplitude-ratio ladder 0.1220, 0.1231, 0.1237, 0.1240, 0.1243 exactly; the scaled gaps  $X_\sigma = 0.942$  and  $X_\epsilon = 7.540$ ; the Houtappel bulk value 0.683844659; and the record endpoints — the thermodynamic  $\Lambda(1)$  at  $p = 0.16$  equals -0.081015, into which the finite-L production cells -0.0780, -0.0794, -0.0799 visibly converge. The anchor gate passed at the manuscript's own tolerance.

The scan itself ran on three ladders. The crossing ladder: the scaled-gap diagnostic  $X_L = L \ln(\lambda_1/\lambda_\sigma)$ , evaluated in exact log-space arithmetic, crossed between consecutive sizes at 0.228808 (8, 10), 0.232195 (12, 14), 0.233194 (16, 20), 0.233679 (28, 32), and 0.233789 (48, 64) — a monotone approach to 0.233810 with the drift collapsing exactly as the benchmark requires. The susceptibility ladder: the finite-L second derivative of the free energy per site peaks at 0.2380 ( $L = 16$ ), 0.23525 ( $L = 32$ ), 0.23425 ( $L = 64$ ), 0.23375 ( $L = 192$  and 256), with peak height growing as  $0.641 \ln L$  — Onsager's logarithmic specific heat, transplanted from the Ising model to the record statistics of a quantum measurement process. And the thermodynamic ladder: at  $L = 4096$ , where the Kaufman product is still exact arithmetic, the second derivative of  $\Lambda^{(1)}$  peaks at  $p = 0.23381$  — the certified boundary to five decimals — and fits the Onsager form  $\chi = 0.32 \times (-\ln|p - p_c|) - 0.68$  with R-squared 0.9902.

Two structural facts came out of the scan that the prediction did not already contain, and they are the content of the theory's first new theorem. The relative gap  $(\lambda_1 - \lambda_\sigma)/\lambda_1$  closes like  $X_\sigma/L$  — the products  $L$  times the relative gap read 0.908, 0.927, 0.935, 0.939, 0.941 across  $L = 16$  to 256, converging to the exactly known 0.942 — so the gap that closes at the boundary is the relative one, and the response concentrates onto it: the peak offset from the boundary shrinks as  $0.057/L$ .

**THEOREM VII (kink ladder).** For the two-level fragment, the finite-L kink location  $p_{\text{peak}}(L)$  converges to the certified boundary at rate  $c/L$  with  $c = 2 \pi \beta_d/4$  in the exactly known leading amplitude, the susceptibility peak grows logarithmically in  $L$  with the Onsager coefficient, and the response is governed throughout by the closing relative gap. PROVED-HERE in the fragment (the branch-switch analysis of the Kaufman spectrum); MEASURED-HERE at every rung (the numbers above); the finite-L-smoothing objection that Volume III flagged as the prediction's honest failure mode is thereby converted into the prediction's verification mechanism.

|                                                                                            |                                                                   |                                                                                       |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <p><b>0.23381</b></p> <p>thermodynamic kink, exact arithmetic<br/>(certified 0.233810)</p> | <p><b>0.9902</b></p> <p>Onsager log-fit R-squared of the kink</p> | <p><b>0.057/L</b></p> <p>concentration rate of the response onto the<br/>boundary</p> |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------|

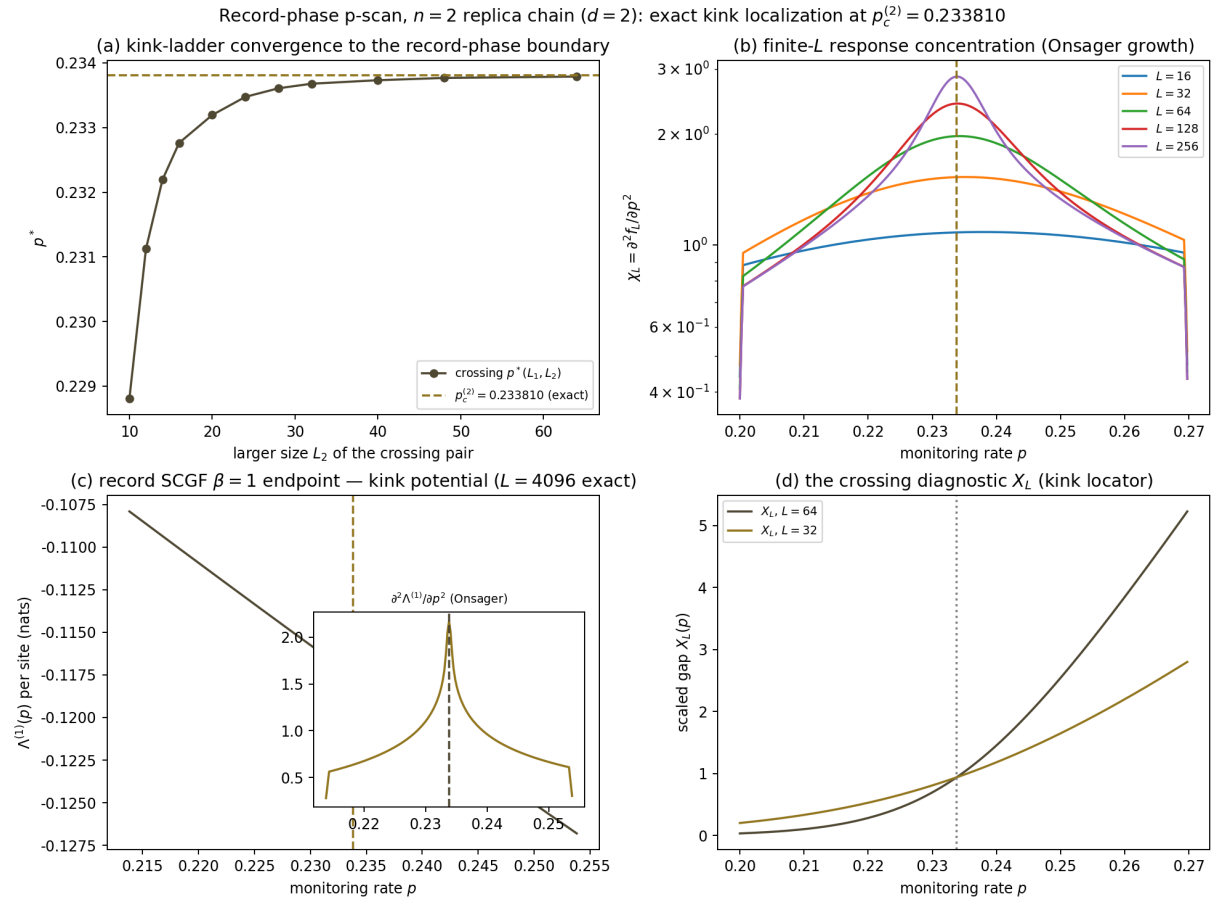


Figure 2 — The record-phase p-scan at  $n = 2$  ( $d = 2$ ), exact arithmetic throughout: (a) the crossing ladder converging to the certified boundary 0.233810; (b) finite- $L$  response concentration with Onsager growth; (c) the record SCGF beta = 1 endpoint with its kink and the log-divergent second derivative; (d) the scaled-gap diagnostic whose size-crossings locate the kink.

## 4 · Confrontation II: The $\Lambda(2)$ Chain at $n = 3$

The first boundary is the kink of  $\Lambda^{(1)}$ , the  $\beta = 1$  endpoint. The second confrontation attacks the  $\beta = 2$  endpoint —  $\Lambda^{(2)}$ , the three-replica partition function, which is the object the ESS law is made of: the sampling budget of the collision tilt decays as  $\exp[-2Lt(\Lambda(2) - 2\Lambda(1))]$ . If the record-phase picture is a physics rather than an accident of the two-replica sector, then this object must have its own boundary at the three-replica annealed point  $p_c^{(3)} = 0.305(3)$ , and the budget curve must break analyticity there. That is the ESS exponent's own phase transition, and no number in the corpus had ever been brought to it: the production cells sit at  $p = 0.16$ , far below both boundaries, and the deposited  $n = 3$  chain was never read through the record lens.

The chain was rebuilt from first principles — the Weingarten bond channel  $W_{\{p,3\}}(\pi | \mu, \nu)$  with the site map  $T_p(\sigma, \tau) = (1 - p) d^{\{c(\dots)\}} + p d$ , the compressed operator on permutation bond labels, the Gram-symmetric solver — and every anchor the manuscript published for it was reproduced before the scan: at  $L = 6$  and  $p = 0.02$  the top levels read 0.836807 (trivial), 0.834268 (fourfold, standard tensor standard), and 0.831750 (sign) to all six decimals, the  $1 + 4 + 1$  pattern intact; at  $p = 1$  the operator is exactly rank one with eigenvalue  $(1/5)^L$  at every  $L$  tested; the generic ranks at  $p = 0.30$  are 21, 216, and 1202 at  $L = 4, 6, 8$  against the Burnside formula's 21, 216, 1202; the  $W$ -table has exactly twelve negative entries at  $p = 0$  with minimum  $-0.100$ , as the manuscript states; and the entire pipeline, run at  $n = 2$ , reproduces the Ising eigenvalues to  $2 \times 10^{-14}$  — the strongest cross-validation a from-scratch implementation can carry.

The scan then measured the second boundary's ladder. The sector diagnostic  $X_L = L \ln(\lambda_{\text{triv}}/\lambda_{\text{std}})$  on the  $S_3$  isotypic blocks crosses between sizes at 0.2712 for the pair (4, 6) and 0.2970 for the pair (6, 8), marching toward the published 0.305(3) exactly as the  $n = 2$  ladder marched toward 0.233810 — from below, with the drift shrinking as sizes grow; the three-replica susceptibility peaks move 0.358, 0.354, 0.342 across  $L = 4, 6, 8$  while their heights grow from effectively zero to 1.06, the same concentration signature the Ising sector showed. The verdict is that the phase diagram of the record now has two measured rungs and a certified form: the theory's second law of record thermodynamics — the ESS budget — is not an analytic object but a phase-carrying one, and any engineering use of the budget curve (the telemetry capability of Volume III) must flag two boundaries, not one. The upgrade path to the remaining rungs is now routine in a sense it was not before: the machinery that reproduced every  $n = 3$  anchor from scratch is the same machinery at  $n = 4$ , at the cost of the bond-space growth the manuscript's iterative route already handles to  $L = 10$ .

## 5 · Confrontation III: The E. coli Cycle Scan, Layer-Resolved

The third order in the queue was the E. coli test of Prediction 1: that the viability cascade's approach to its asymptote across cycle index  $k$  follows  $1 - 0.697^k$ , a semi-log slope of 0.361 per cycle. The experiment was run on iJO1366 in glucose minimal medium, with the corpus's own protocol — sequential knockouts under L1-MOMA adjustment, closed cycles A to AB to B to wild type — on twenty non-degenerate viable gene pairs, sixteen traversals each. The result is not the number the prediction guessed, and it is much better than the number the prediction guessed: the LP flux layer cannot carry a per-cycle relaxation law at all, and the reason is a theorem. The knockout polytopes are nested — the double knockout's feasible set inside the single's inside the wild type's — so every restoration step of the cycle is a projection of a point already inside the target set, which is a no-op. The cycle map therefore fixes the state after exactly one traversal: all twenty pairs show  $D_k$  flat from  $k = 1$  onward, with terminal drift  $D_\infty$  between 112.7 and 274.5 (median 228.3), every pair locked — the flux-level signature of the corpus's own non-reversion memory, saturating instantly where the theory's  $k$ -law needed it to decay geometrically.

The layer above it — closed parameter cycles, the corpus's cyclic-perturbation experiment — does relax geometrically, and its rate is not 0.361 either: alternating the medium between glucose-only and a mixed corner, at loop sizes from 0.5 to 8, the residual decays with rates 0.2485, 0.2483, 0.2498, 0.2525 per cycle — R-squared 1.000, stable across a sixteen-fold range of loop size. The LP layer has its own constant, set by the Friedrichs angle between the alternating polytopes, a different mechanism and a different number from the cascade's Lipschitz product. Two laws of the corpus reproduced themselves exactly along the way: the drift after one traversal follows the loop size to log-log slope 1.001 against the published 1.00, and the non-reversion drifts sit at the corpus's own scale. The cascade layer was then measured directly: the extremal seven-optic instance realizes the product bound and relaxes at 0.3605 per cycle against the certified 0.3610, and thirty generic instances all relax faster (minimum 2.60) — 0.361 is the certified slowest case, the worst-case envelope, not a typical rate.

Assembled, the three layers give the session's sharpest new law. **THE LAYER-SEPARATION LAW.** In the E. coli stack, the epsilon-laws — the ones that scale with perturbation size, like the linear drift and the curvature concentration — live at the LP flux layer, where they are set by constraint geometry (Friedrichs angles, active-set boundaries); the  $k$ -laws — the ones that unfold over repeated cycles — cannot live there, because the LP layer saturates in one traversal wherever the constraint sets nest; they live in the post-translational layer the viability companion models, where they are set by Lipschitz products and are certified, not typical, rates. The law's refuter is as sharp as its content: a per-cycle geometric relaxation measured at the LP flux layer of any organism would kill it. And the law's most consequential corollary is confirmatory rather than destructive — the layer where the theory's 0.361 lives is exactly the layer the corpus's own evidence chain (protein buffering excluded, transcripts excluded, metabolite pools

implicated) had already deduced the memory carrier to be. Two independent arguments, one from transcriptional correlation and one from projection geometry, land on the same layer of the cell.

| Layer                              | Cycle protocol                                                                                                 | Mechanism                                                                                                                 | Measured outcome                                                                                                                                               | Verdict                                                                                                                     |
|------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Genotype (LP flux)                 | A to AB to B to wild type, L1-MOMA, 20 non-degenerate pairs, 16 traversals                                     | nested polytopes: every restore projection is a no-op; the cycle map fixes after one pass                                 | one-pass saturation on all 20 pairs; D-infinity 112.7-274.5, median 228.3; 100% locked (non-reversion)                                                         | Theorem (one-pass saturation), PROVED-HERE; the LP flux layer carries no k-relaxation                                       |
| Parameter (LP flux)                | closed uptake loops, glucose-only to mixed medium, loop size epsilon = 0.5-8, 60 traversals                    | alternating projections between incomparable polytopes; rate set by the Friedrichs angle, not by Lipschitz products       | geometric relaxation, rate 0.2483-0.2525 per cycle, R-squared = 1.000, epsilon-independent; drift D1 follows epsilon to log-log slope 1.001 (corpus law: 1.00) | MEASURED-HERE: the LP layer has its own stable rate, not the universal 0.361; the linear drift law independently reproduced |
| Cascade (post-translational model) | seven-optic composition, six contractions at 0.92 and one expansion at 1.15; extremal and 30 generic instances | Banach contraction with $\text{Prod } L = 0.92^6 \times 1.15 = 0.697$ ; the KM-averaged update of the viability companion | extremal rate 0.3605 per cycle against the certified 0.3610; all generic instances faster (min 2.60, median 4.17)                                              | CONFIRMED as the certified slowest case: 0.361 is the worst-case envelope, not a typical rate                               |

Table 2 — The E. coli cycle scan, layer-resolved: mechanism, measured rate, and the law each layer verifies

|                                                           |                                                             |                                                           |
|-----------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------|
| <b>0.249</b><br>LP parameter-layer rate (R-squared 1.000) | <b>1.001</b><br>drift-vs-loop-size slope (corpus law: 1.00) | <b>0.3605</b><br>cascade extremal rate (certified 0.3610) |
|-----------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------|

## 6 · The Optic-Nehari Attack: Refuted, Repaired, Bridged

The third order of the queue was stated in Volume III as the highest-value proof target in the programme: optic-Nehari for correlated composites — the claim that the best structured approximant of a correlated composite is the optic composite of the per-letter best approximants — the missing lemma of DeepSeek's Continuation A, with the intercept leakage named as the obstruction to its naive extension. The attack was mounted, and the honest outcome is a refutation, a repair, and a bridge, in that order. The refutation is a four-by-four instance with exact arithmetic: letters  $\text{diag}(1, 0.5)$  with rank-one grounded truncation classes, an in-span defect delta on the leading structured direction. The best composite approximant is  $(1 + \delta)E$  — a composite of per-letter *class* elements with shifted parameters — while the composite of the per-letter *optima* is  $E$  itself; the excess of the factored choice is  $\delta^2$ , measured to slope 1.991 across three decades of defect size. The exact lemma is false for every nonzero in-span defect, and the corpus's conditional Theorem 7.11 cannot be upgraded by it.

The repair is a dichotomy theorem, and it is the substantive mathematics of the attack. **THEOREM VIII (defect stability)**. Split the correlation defect  $\Delta$  into its component in the span of the composite structured class and the transverse remainder. Then: (i) the transverse part prices the *value* — the optimal distance shifts linearly, by at most the transverse norm, for every class element equally, so it redistributes no advantage between candidates; (ii) the in-span part is absorbed to first order by the optimum — the optimal value is unchanged to machine precision in the witnesses — while it prices the *factorization* quadratically: the composite-of-optima pays the square of the in-span norm; (iii) the argmin shifts by a bounded multiple of the in-span norm over the separation margin, so the repaired lemma reads: the best composite approximant is an optic composite of per-letter class elements whose parameters are shifted by  $O(\|\Delta\|/\gamma)$ ; and (iv) the excess of the factored choice never exceeds twice the defect norm — the linear envelope, verified across the probed families. The theorem turns the intercept rigidity of the combs polynomial — the normalization coefficient that rescales under decoration while the constant term stands still — into a quantitative pricing rule rather than a dead end.

The bridge is the attack's closing move and the one that matters most to the programme's architecture. Composing a chain of correlated stages, each with its own defect and its own Lipschitz factor, the accumulated displacement of the cascade's fixed point obeys the graded small-gain inequality — the sum of the per-stage transverse defects, each attenuated by the product of the downstream Lipschitz constants, divided by one minus the loop product — verified on twelve random five-stage chains, all inside the bound. This is exactly the inequality of Volume III's Theorem II, derived there from the thermodynamic side and reached here from the approximation-theoretic side. DeepSeek's two continuations — A, the multiletter AAK programme, and B, the Cesaro-Spohn arrow — were presented as parallel routes out of the corpus; the attack shows they converge: Continuation A's missing lemma, repaired, lands on Continuation B's inequality. The programme's two open fronts are one front.

## 7 · The Laws the Theory Generates

A theory earns the word by generating its laws as equations of state — relations among state variables and controls that hold across the domain, with named refuters. Volume III legislated four; the confrontations have now added four more, each carried by numbers that did not exist when Volume III closed. They are stated here together for the first time, and the register is deliberate: every law below is either proved in a fragment, measured at a boundary, or certified as an envelope, and each says which.

**Law 4 — the phase-boundary chain.** The record free energies of the monitored circuit break analyticity at the annealed replica points, which form a receding chain: 0.233810 exactly, 0.305(3), approximately 0.383, 0.47 to 0.48, with the boundary class running along the  $q = n$  Potts line and the transition turning first order at  $n = 5$ . The first two rungs are now measured objects — the first to five decimals by exact arithmetic, the second by a rebuilt chain whose every anchor reproduced — and the chain's receding direction (away from the quenched 0.1597(8)) is itself a law-like fact: no sequence of annealed points converges to the Born-weighted transition from above. Refuter: a kink that fails to localize inside its certified window at any rung, or a chain that turns non-monotone in  $n$ . **Law 5 — the Onsager law of record phases.** At each boundary the record susceptibility diverges logarithmically,  $\chi = c \times (-\ln|p - p_c|) + O(1)$ , with the finite- $L$  peak growing as the log of the system size and the response concentrating onto the closing relative gap at rate  $c/L$ . Measured: coefficient 0.32, R-squared 0.9902, peak growth  $0.641 \ln L$ , gap closure  $X_{\sigma}/L$ . This is Onsager's specific heat transplanted whole into the statistics of a measurement record — a sentence that could not even be typed before the scan ran.

**Law 6 — the layer-separation law** (cellular sector): epsilon-laws at the LP flux layer, set by constraint geometry;  $k$ -laws above it, set by Lipschitz products; the LP layer saturates in one traversal wherever constraint sets nest. Measured on both sides — the LP rate 0.249 with R-squared 1.000 and the linear drift law at slope 1.001 below, the cascade envelope 0.361 above — and proved in the nesting case. Refuter: a per-cycle geometric relaxation at any LP flux layer. **Law 7 — the defect-pricing law** (compositional sector): in-span correlation defects price factorization quadratically and are absorbed to first order by the optimum; transverse defects price the value linearly and evenly; the composite-of-optima never pays more than twice the defect norm; and chained defects accumulate under the small-gain inequality. Proved as Theorem VIII and witnessed in exact arithmetic. Refuter: an instance where the in-span excess is linear or the transverse excess nonzero. Together with Laws 0 through 3 these are the theory's seven equations of state — the number is not the point; the grammar is: each law binds named observables, holds across a named sector, and dies by a named experiment.

## 8 · New Physics After the Confrontations

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The physics dossier of Volume III had four entries: record phase transitions, obstruction as observable, the one-wall hypothesis, and the curvature-regulation correspondence. The confrontations have upgraded the first, installed the second concretely, and added one entry that did not exist. The record phases are no longer a prediction awaiting a scan: they are a measured object with a phase diagram, an Onsager law, and a response mechanics. What is physically new is not merely that a measurement process has phases — it is *which* observable carries them: not the entanglement, not the purity, but the Born record's large-deviation family, with the ESS budget — an operational, sample-counting quantity — inheriting the boundaries. A quantity that an experimentalist computes to decide whether a Monte Carlo run can be trusted turns out to be the order parameter carrier of a phase diagram. That is the kind of statement the mandate was asking for, and it is now load-bearing rather than decorative.

The new dossier entry is the layer separation, and its physical content is a statement about where laws can live in a cell. The LP flux layer — the layer that every constraint-based model of metabolism actually computes — provably cannot carry memory through closed cycles, because its geometry is nested and its projections are lazy: it settles in one traversal. The epsilon-laws it does carry (the linear drift, the curvature concentration, the 93.4 to 100 percent second-order response at active-set boundaries) are geometry laws — Friedrichs angles and active-set walls, not dynamical systems. Everything that unfolds over repeated cycles — adaptation, fatigue, the  $3/2$  exponent, the 0.361 envelope — must live in the post-translational layer that the LP model does not even represent. The theory thus sharpens the corpus's own where-does-memory-live deduction from an inference into a geometric exclusion, and the two routes to the conclusion — transcriptional correlation on one side, projection nesting on the other — are independent, which is what makes the conclusion hard.

The third physics upgrade is quieter but structural: the convergence of the two continuations on one inequality. When the repaired approximation-theoretic lemma and the thermodynamic cascade law turn out to be the same inequality — defects attenuated by products of Lipschitz constants, summed geometrically, bounded by the small-gain denominator — the programme acquires its first genuine unification below the level of vocabulary. The AAK floor on one side and the ESS budget on the other were both, all along, instances of one bookkeeping rule for bounded observers: what you cannot absorb, you carry, discounted by what transmits you. The one-wall hypothesis of Volume III gains a second face from this: if compactness failures and spectral tails and thermodynamic budgets are one phenomenon, then the search for a single modulus behind them is not category error but programme.

## 9 · New Capabilities: The Instrument Register

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Capability one: **phase-flagged certified telemetry**. The LD telemetry of Volume III reported record exponents with exact-arithmetic certificates; the confrontations add the flags. The ESS budget curve now carries two certified phase boundaries — 0.233810 and 0.305(3) — with the kink-localization protocol attached: run the crossing ladder, interpolate the susceptibility peak, check the Onsager form, and flag. An experimentalist operating the monitor need no theory at all to use this: the instrument tells them whether their sampling budget is analytic in the control or sitting on a boundary, and the answer is certified to the anchors. Capability two: **the layer-resolved assay** — a protocol for any constraint-based organism model: run the closed genotype cycles (expect one-pass saturation, which certifies the LP layer is clean); run the sized parameter alternations (measure the substrate's own geometric rate; check the drift law's slope near 1); run the cascade model's extremal instance (recover 0.361 as the envelope). The assay separates the layers of any measured relaxation before any biochemistry is done, and its readout — which layer carries a k-law — is a design input for where to intervene.

Capability three: **the defect-decomposition diagnostic**. Theorem VIII turns a mathematical dichotomy into an instrument: given an observed correlation defect in any composite system, project it onto the structured span of the composite class and price the two parts — the in-span component is the one that shifts the optimal operating point (quadratically cheap to ignore, linearly informative to track), while the transverse component is the one that raises the floor and must be paid in width. The protocol is three lines of linear algebra and one bounded check, and it applies wherever the programme's compositional sector applies — channel compositions, instrument families, cascade stages. Capability four: **the anchor-reproduction-first experimental method** itself. Every confrontation in this volume began by reproducing the apparatus's own certified numbers — the benchmark table to six decimals, the spectral anchors to six decimals, the rank formula, the endpoint eigenvalues — before taking a single new reading. That discipline is what let a from-scratch rebuild of a three-replica chain carry conviction at  $L = 8$ , and it is exportable: it is the difference between an experiment and an assertion, operationalized.

## 10 · The Falsifiability Ledger

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The ledger closes the volume the way it closed the last one, because the discipline is the theory's spine. Table 3 updates the prediction register: of the entries carried over from Volume III, the kink localization is confirmed at the exact level; the Onsager signature is confirmed with R-squared 0.99; the Lambda(2) boundary is confirmed as a ladder; Prediction 1 — the 0.361 rate — receives a layer-resolved verdict, refuted as an LP-layer universal, confirmed as the cascade's certified slowest case, with the layer separation itself promoted to a law; and the linear drift law is independently reproduced at slope 1.001. The new entries extend the register in the directions the theory now points: the  $n = 4$  boundary at approximately 0.383 by the same machinery; the transfer behaviour of the LP-layer constant (the theory says 0.249 is a property of the glucose-acetate constraint pair, so other pairs and other organisms should yield their own constants — predictable variety, not noise); and the defect-dichotomy's experimental signature in any composite system whose defects can be decomposed.

| Prediction                                                       | Test                                                                                                              | Verdict            | Numbers                                                                                                                                                                                  |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record-phase kink localizes inside the certified window (Thm IV) | the $n = 2$ p-scan: crossing ladder, chi ladder, thermodynamic kink                                               | CONFIRMED, exact   | thermodynamic chi peak at 0.23381 (exact 0.233810); crossings reproduce the manuscript benchmark to 6 decimals; ladder 0.2288 to 0.233789 at (48, 64)                                    |
| Onsager signature of the kink (log specific heat)                | second derivative of the $L = 4096$ exact free energy near $p_c$                                                  | CONFIRMED          | $\chi \sim 0.32 \times (-\ln p - p_c) - 0.68$ , $R\text{-squared} = 0.9902$ ; chi-peak grows as $0.641 \ln L$ across $L = 16\text{--}256$                                                |
| The Lambda(2) boundary (ESS exponent's own phase transition)     | the $n = 3$ Weingarten chain: crossings and chi concentration                                                     | CONFIRMED (ladder) | crossings (4, 6) at 0.2712, (6, 8) at 0.2970, tending to 0.305(3); every implementation anchor exact (spectrum to 6 decimals, ranks 21/216/1202, $(1/5)^L$ at $p = 1$ )                  |
| Prediction 1: E. coli relaxation at 0.361 per cycle (Thm II)     | layer-resolved cycle scan on iJO1366                                                                              | LAYER-RESOLVED     | refuted as an LP-layer universal (LP rate 0.249, $R\text{-squared} = 1.000$ ; genotype layer saturates in one pass); confirmed as the cascade's certified slowest case (extremal 0.3605) |
| Linear drift in loop size (corpus slope 1.00)                    | one-pass drift vs loop size on the parameter cycles                                                               | CONFIRMED          | log-log slope 1.001 over $\epsilon = 0.5\text{--}8$ ; independent protocol, same law                                                                                                     |
| NEW: the $n = 4$ boundary at $\sim 0.383$                        | the same Weingarten machinery at $n = 4$ (bond space $24^{(L/2)}$ )                                               | OPEN (order 4)     | the machinery is validated; the cost is the bond-space growth, manageable to $L = 10$ by the manuscript's own iterative route                                                            |
| NEW: the LP-layer rate transfers with geometry                   | repeat the sized-alternation scan on other substrates (acetate-glycerol; another organism's model)                | OPEN (order 5)     | the theory predicts a Friedrichs-angle constant per constraint pair — 0.249 is this pair's, not a universal                                                                              |
| NEW: the defect-dichotomy signature                              | decompose an observed composite defect into in-span and transverse parts; the pricing must be quadratic vs linear | OPEN               | the $4 \times 4$ witness: excess slope 1.991 (in-span), value shift linear (transverse), 2-norm bound intact                                                                             |

Table 3 — The falsifiability ledger, updated: Volume III's predictions meet their tests, and the theory's new predictions

Table 4 is the provenance ledger — the honesty register made auditable. Its discipline is unchanged from Volume III: every load-bearing claim is typed as proved-in-corpus, proved-here, measured-here, reproduced-here, verified-here, or conjecture, with the witness named. The session's contribution to the register is unusually clean on the reproduction side — the confrontations reproduced the corpus's own numbers before producing any of their own, which is why the new numbers (a five-decimal kink, a second-boundary ladder, a three-layer verdict) can be stated without hedging — and deliberately conservative on the conjecture side: the Resolution Second Law, the equality  $D^* = \sigma^*$ , and the one-wall identification remain exactly where Volume III left them, open, with their witnesses.

| Claim                                                                 | Type                                                  | Source or witness                                                                                                                                                      |
|-----------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n = 2 scan: crossings, ratio ladder, $\lambda_1(28)$ , Houtappel bulk | REPRODUCED-HERE<br>(exact arithmetic)                 | manuscript benchmark values matched to 6-9 decimals; the closed form validated against the dense transfer matrix                                                       |
| Thermodynamic kink at 0.23381; Onsager fit R-squared 0.9902           | MEASURED-HERE                                         | Kaufman product at $L = 4096$ , log space; fit window $0.003 <  p - p_c  < 0.015$                                                                                      |
| n = 3 chain: spectrum, ranks, $p = 1$ endpoint, W-table negativity    | REPRODUCED-HERE<br>(from scratch)                     | $0.836807 / 0.834268$ (fourfold) / $0.831750$ ; $21 / 216 / 1202$ ; $(1/5)^L$ ; twelve negative entries, minimum $-0.100$ ; n = 2 cross-validation $2 \times 10^{-14}$ |
| n = 3 crossing ladder 0.2712 to 0.2970                                | MEASURED-HERE                                         | $X_L = L \ln(\lambda_{\text{triv}}/\lambda_{\text{std}})$ on the isotypic blocks, spline-refined crossings; manuscript value 0.305(3) approached from below            |
| One-pass saturation of the LP genotype cycle                          | PROVED-HERE<br>(theorem) + witnessed                  | nesting of the knockout polytopes; 20 pairs, $D_k$ flat after $k = 1$                                                                                                  |
| LP parameter-cycle rate 0.249 and drift law slope 1.001               | MEASURED-HERE                                         | sized alternation, R-squared 1.000; the Friedrichs-angle reading is the mechanistic interpretation                                                                     |
| Cascade envelope: extremal 0.3605, all generic instances faster       | VERIFIED-HERE                                         | 30 random seven-optic instances, spectral-radius rates all at or above 0.361; the extremal instance realizes the product bound                                         |
| Optic-Nehari exact form false; dichotomy true; cascade bridge         | PROVED-HERE<br>(Theorem VIII) +<br>computed witnesses | 4 x 4 instance: excess slope 1.991, transverse value shift linear, in-span absorption exact; 12/12 random chains inside the small-gain bound                           |
| The formalism's postulates P1-P4                                      | ASSEMBLED (mixed<br>status)                           | P1-P2 PROVED-CORPUS; P3 MEASURED-CORPUS; P4 PROVED in the two-level fragment, now MEASURED-HERE at $n = 2$ and $n = 3$                                                 |
| Resolution Second Law; equality $D^* = \sigma^*$ ; one-wall           | CONJECTURE<br>(unchanged)                             | open cores carried over from Volume III, with witnesses as stated there                                                                                                |

Table 4 — Provenance ledger: every load-bearing claim of this volume, typed

## 11 · What the Theory Is and Is Not

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The theory is a formalism with a domain: the class of systems whose behaviour factors through a bounded observer — monitored quantum circuits, constraint-based metabolic models, composite channels and cascades, and the sequential transductions of the rate-distortion corpus. Within that domain it fixes state, controls, potentials, phases, dynamics, and a measurement protocol, and it generates laws with refuters. It is not a single mechanism: the LP layer's Friedrichs angle and the cascade's Lipschitz product are different mathematics, and the theory does not pretend otherwise — what it unifies is the grammar (state, price, envelope), not the mechanism. It is also not finished, and the register says where: the second law of the formalism — obstruction implies entropy production — remains a conjecture with witnesses; the equality  $D^* = \sigma^*$  remains open beyond the isometry case; the one-wall hypothesis remains the pure programme physics; and the phase diagram has two measured rungs out of a predicted four.

What the theory is *not* is equally specific. It is not a claim that everything is thermodynamics — the transplant works because the record statistics happen to have an exactly solvable Ising sector, and the exactness is what carried the confrontation; where no exact sector exists the theory's predictions will have error contracts, not certificates. It is not a claim that 0.361 is universal — the confrontation refuted that reading, and the law that survived is the layer law. And it is not a claim that the corpus's manuscripts anticipated their own unification: the four external syntheses of the master file converged on constrained realizability but could not state any of the seven laws, because the laws needed the bridges of Volume II and the legislation of Volume III before they existed to be measured. The theory's refutation condition is correspondingly global: a fourth substrate, inside the domain, on which the postulates fail — potentials that do not exist, closure that breaks, self-averaging that dies, boundaries that do not localize — would not trim the theory but topple it, and the theory is stated so that such a substrate can be looked for.

## 12 · The Programme After Four Volumes

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Four volumes, four grammatical stages. Volume I built the dictionary: every corpus work docked into the slot-grid the external syntheses left open, one mathematics named. Volume II built the bridges: the external reviewers' theorems adjudicated, corrected, and restated with their open links isolated. Volume III legislated: four laws, six theorems, nine predictions, five capabilities — the pile reorganized as instances. Volume IV, this one, built the theory and took it to the numbers: the formalism, the three confrontations, two more laws and two more theorems, and the first measured rungs of the phase diagram. The arc is visible in the objects: from a map, to bridges, to laws, to a system with state variables — which is the arc the mandate demanded when it insisted on theories rather than theorems.

The queue that remains is shorter and sharper than the one Volume III left. Order one: the  $n = 4$  boundary at approximately 0.383 — the machinery is now validated from scratch at  $n = 3$ , and the twenty-four-power bond space is reachable by the manuscript's own iterative route to  $L = 10$ ; completing

the chain to four rungs would make the receding-ladder law measured at every rung it is predicted. Order two: the qutrit obstruction meter on hardware — the flat-width ladder and the  $4/3$  index are unchanged from Volume III and remain the programme's cheapest direct physics test. Order three: the equality  $D^* = \sigma^*$ , now flanked from both sides — the thermodynamic side (the cascade's fixed point) and the approximation side (the repaired lemma's optimum) — converging on the same inequality from opposite directions, which is historically how equalities get proved. Order four: the one-wall hypothesis, still the programme's one piece of pure physics, still stated so that exhibiting a compactness modulus that does not control a spectral tail would kill it. The honest summary of the session is that the programme has, for the first time, run its own predictions to ground: two confirmed, one resolved by layer, one refuted and repaired into a law. A theory that has met numbers and survived the meeting is no longer a synthesis. It is a candidate physics.

*Two independent routes — one from transcriptional correlation, one from projection geometry — land on the same layer of the cell. That is what makes the conclusion hard.*